

ChainBuddy: An AI-assisted Agent System for Helping Users Set up LLM Pipelines

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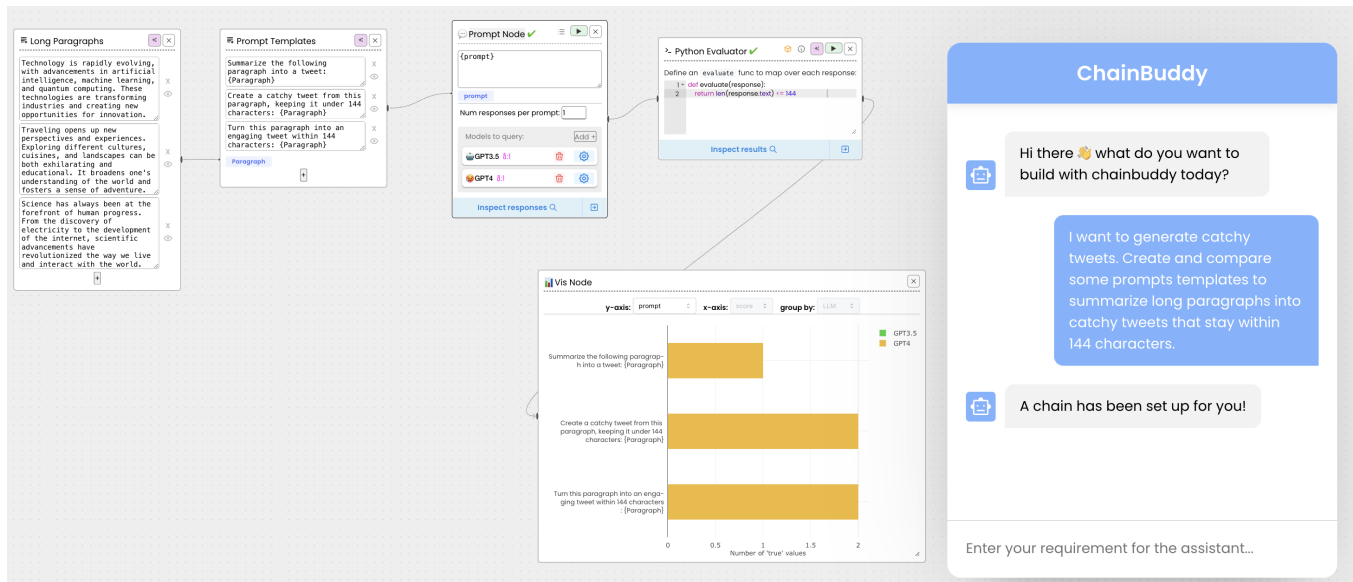


Figure 1: An example workflow created by ChainBuddy.

ABSTRACT

As large language models (LLMs) continue to advance, their potential applications have grown significantly. However, one persistent challenge remains: evaluating LLM behavior and crafting effective prompt chains. Many users struggle with where to start, often referred to as the "blank page problem." ChainBuddy, a new evaluation assistant built into the ChainForge platform, aims to tackle this issue. It offers a straightforward and user-friendly way to plan and evaluate LLM behavior, making the process less daunting and more accessible.

CCS CONCEPTS

- **Human-centered computing** → **User interface design.**

KEYWORDS

large language models, agentic system, intelligent user interfaces

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1 INTRODUCTION

Over the past year, the growing interest in AI has spawned a plethora of tools, APIs, and best practices for creating applications based on LLMs. These advancements include a wide range of techniques, from prompt engineering and LLM evaluation platforms to sophisticated AI agent systems equipped with tool use capabilities.

Despite these advancements, many users encounter a significant challenge: the "blank page problem." This problem is characterized by the uncertainty and difficulty of knowing where to begin when using platforms like promptfoo [5] or Flowise [2]. Some researchers have proposed social solutions to this problem, such as the development of prompt repositories and community-sharing mechanisms [7]. Another proposed solution is to create an assistant that helps users get started by providing initial guidance and structure [1].

Taking the latter approach, our solution, ChainBuddy, is an AI-powered assistant that automatically generates starter LLM workflows (“flows”) given an initial prompt. ChainBuddy help users get started with evaluating LLM behavior and setting up chains by

providing a starter flow, custom-tailored to their use case, that they can edit and extend. We built ChainBuddy on top of an existing open-source visual environment, ChainForge, which was designed for prompt engineering, LLM evaluation, and experimentation [1]. ChainForge allows users to query multiple LLMs, compare response quality, and establish evaluation metrics with minimal coding required. By building on top of this foundation, ChainBuddy offers users structured assistance to help them overcome the blank page problem, making it easier to explore, experiment and evaluate the behavior of LLMs across a wide range of potential tasks and use cases.

2 DESIGN

When designing ChainBuddy, we focused on creating a flexible, user-friendly interface that could support a wide range of use cases beyond just prompt engineering. To achieve this, we decided on a chatbot-style interface within the ChainForge platform. This choice was driven by the need for an intuitive and interactive environment that can support various users based on prior work of ChainForge such as data processing, prompt optimization, LLM auditing, and more. Our goal was to ensure that ChainBuddy could handle diverse and complex requirements while being accessible to users with varying levels of technical expertise and use cases. Note that our ultimate goal is to extend ChainBuddy to guide users beyond a single-turn interaction, i.e., to edit existing flows; however, we consider here only the ability of ChainBuddy to generate flows from scratch.

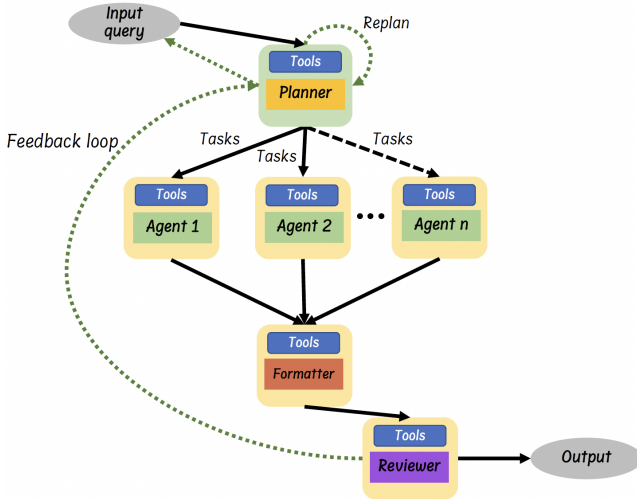


Figure 2: System architecture of ChainBuddy

2.1 Architecture

ChainBuddy is built on LangGraph [3], a library designed for constructing stateful, multi-actor applications with LLMs. LangGraph’s core benefits include its ability to handle cycles, provide fine-grained controllability, and ensure persistence. These features are essential for creating reliable agent-based workflows that can support advanced human-in-the-loop and memory functionalities.

Inspired by concepts from *Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models* [6] and projects like Baby-AGI [4], we designed ChainBuddy’s agentic system to generate long-term plans based on user requirements. This approach involves breaking down each task into specific, manageable actions that can be executed by individual agents. This design allows each agent to focus on a single task, improving efficiency and accuracy.

Key architectural features include:

- **Explicit Long-Term Planning:** By generating comprehensive plans at the outset, it ensures clarity and direction throughout the evaluation process.
- **Task-Specific Agents:** Each task in the plan is assigned to a specific agent, allowing for focused execution. This specialization helps in utilizing smaller, less powerful models for execution tasks while reserving larger, more capable models for planning.
- **Capacity for Flexible Interactions:** The system supports long-term interactions, enabling users to iteratively refine their prompts and evaluation strategies based on ongoing feedback and results.

By leveraging these architectural principles, ChainBuddy provides a robust, scalable solution for a wide range of LLM-related tasks, making it an invaluable tool for researchers and developers working with ChainForge.

3 EARLY FEEDBACK

We conducted informal pilot studies to gather early feedback and improve our system. Here are some key insights we discovered:

- **Initial Setup and User Guidance:** Users appreciated structured guidance when setting up their evaluation workflows. For instance, a user found it difficult to visualize the kind of results they should expect and suggested including examples and default text to help formulate specific questions. Similarly, another user highlighted the need for clearer instructions and the ability to handle template nodes more intuitively.
- **System Responsiveness and Usability:** Feedback consistently pointed out the importance of system responsiveness. Users suggested incorporating features like visualizing the loading progress, as well as providing explainable AI elements to help users understand how the system arrives at certain visualizations or results.
- **Desire for Editing Existing Flows:** Users found value in ChainBuddy’s ability to handle diverse tasks, from generating prompt chains to complex LLM evaluations. However, some users wanted to retry the generation or edit an existing flow once generated.
- **User Experience and Interaction:** The chatbot interface was generally well-received, but users occasionally perceived it as too similar to a traditional Q&A format, potentially limiting its perceived capabilities. Some users suggested that incorporating more interactive elements and examples would help clarify the system’s potential and guide users more effectively through the setup process.

These insights have been instrumental in refining ChainBuddy, making it more intuitive, responsive, and capable of handling a broader range of tasks.

4 FUTURE WORK

Addressing Potential Biases. One of the key areas for future work involves addressing potential biases within the system. As with any AI-driven tool, there is a risk of inherent biases in the responses generated by LLMs. We plan to implement more robust bias detection and mitigation strategies to ensure that ChainBuddy provides fair and balanced outputs. This includes allowing users to review and provide feedback to minimize bias.

Enhancing Interactivity and Data Import. We aim to enhance its chat-like interface. This includes allowing users to import their own data or other contextual information directly into the system. By supporting data imports, users can leverage existing datasets or relevant context, making the tool more versatile and applicable to a wider range of scenarios.

Supporting Editing of Existing Chains Another important enhancement is enabling users to edit existing chains. Currently, ChainBuddy supports the creation of new chains but lacks the functionality to modify them once created. By allowing users to edit and refine their chains, we can provide a more flexible and dynamic tool that adapts to evolving user needs and preferences.

Providing Fine-Grained User Control Providing users with more fine-grained control over the decision-making process of the agents is a priority. This involves developing features that allow users to influence and guide the agents' decisions more directly, ensuring that the outcomes align closely with their expectations and requirements. Enhanced control mechanisms will empower users to tailor the behavior of the agents to specific tasks and contexts.

Offering Variety in Suggested Chains Finally, we plan to increase the variety of suggested chains provided by ChainBuddy. By offering a broader range of pre-configured chains and templates, users will have more options to choose from, catering to diverse use cases and preferences. This variety will help users find more relevant and effective starting points for their specific needs, further reducing the "blank page problem."

By addressing these areas, we aim to make ChainBuddy a more powerful, flexible, and user-friendly tool for LLM evaluation and prompt engineering. These enhancements will help users leverage the full potential of ChainForge, driving more effective and efficient AI-driven workflows.

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